

As soon as you power the circuit, the line follower should start working and should be following the line. Note that two LEDs have been added in parallel to output of the Opamps. They glow when any sensor detects the black line. When the left sensor detects black surface under it, the left LED glows to indicate that there is a left turn ahead. The robot turns left.

Similarly, when the right sensor detects black colour, right LED glows to show that there is right turn ahead. The robot then turns right.

None of the LEDs glow when there is straight line ahead and the robot keeps on moving straight. We built this robot on PCB also and here is what the circuit will look like when you solder it on a PCB.

Here, we used PIC 18F4220 instead of PIC16F84. PIC 18F4220 is more advanced than PIC16F84 and we used it so that we have some scope for better algorithm and logic, since we plan to make a more efficient line follower in future.

We also built the body using wood instead of cardboard and used four motors instead of two so that it could have more power and ability to climb inclined surfaces too. 

Video!

On this month's DVD, we've included an exciting video of our robot in the making. You'll find the step-by-step instructions easy to follow, especially if you are not familiar with electronics and programming. Write to editor@thinkdigit.com and tell us about your experience.